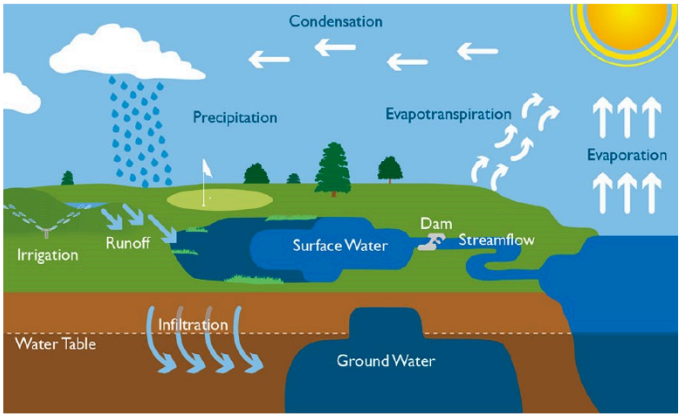
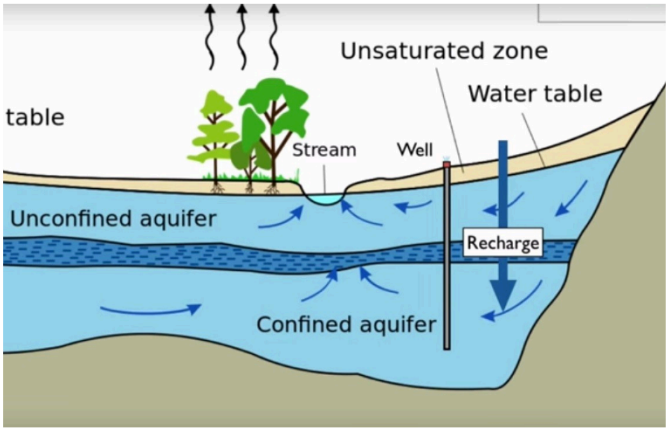
Water Resource

Hydrological Cycle



Water Resource

Fresh water



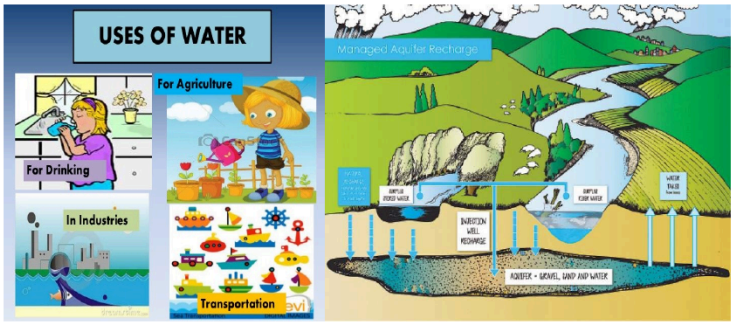
Water Resource

Water Use
Agriculture – 70%
Industry – 22%
Domestic – 8%

Water Storage Ground water – Aquifer

Helps!!
Potable water to drink
Reduce, Reuse, and Recycle (3R)

Concern!!
Ground water depletion
Untreated ground water



Water Resource

Categories



Point



Groundwater



Non-point

Water Resource

Water pollution

Causes

Sewage
Nutrients
Waste water
Chemical waste
Radioactive waste
Oil pollution



Impacts

Death of aquatic (water) animals
Disruption of food-chains
Diseases
Destruction of ecosystems



Water Resource

Water quality

"**Water quality**" is a term used here to express the suitability of water to sustain various uses or processes. Any particular use will have certain requirements for the physical, chemical or biological characteristics of water (UNEP/WHO, 1996)

Water quality – parameters:

- **Physical parameters**. Colour, odour, turbidity, taste, temperature and electrical conductivity
- **Chemical parameters.** the amount of carbonates, sulphates, chlorides, fluorides, nitrates, and metal ions
- **Biological parameters.** Quantity Existence of algae, fungi, viruses, protozoa and bacteria

Water Resource

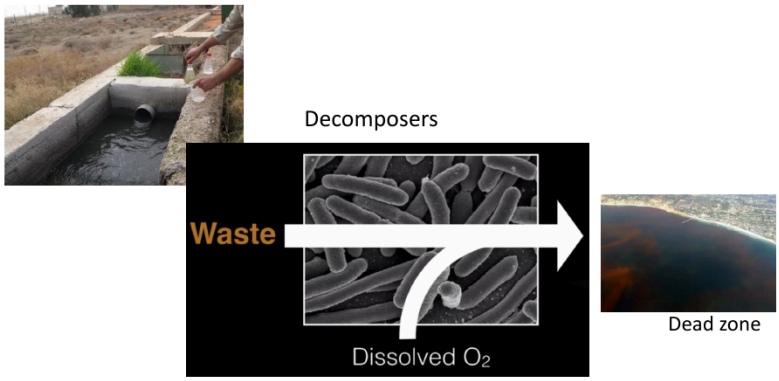
Biochemical Oxygen Demand (BOD)

BOD – the amount of oxygen a quantity of water uses over a period of time at a specific temperature

- Lower BOD values indicate the water is less polluted and higher BOD values indicate it is more polluted by wastewater
- When bodies of water have high oxygen demand due to microbial decomposition, the amount of O_2 for other organisms can be very low

Water Resource

Wastewater



Water Resource

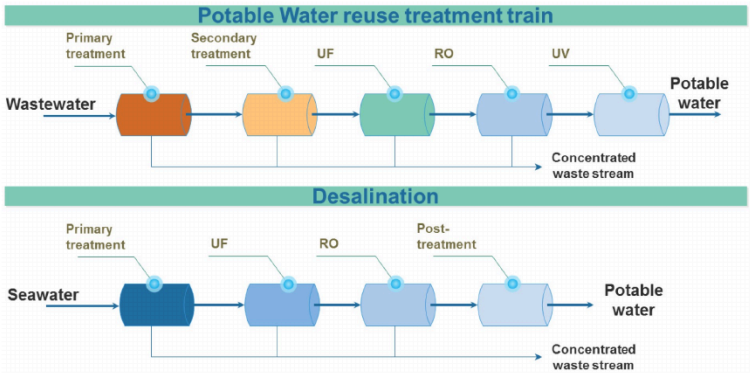
Eutrophication

Eutrophication is an abundance of fertility to a body of water. For instance, Algae bloom

- caused by an increase in nutrients (including nitrogen and phosphorus), such as fertilizers.
- can cause a rapid growth of algae which eventually dies, causing the microbes to increase the BOD.

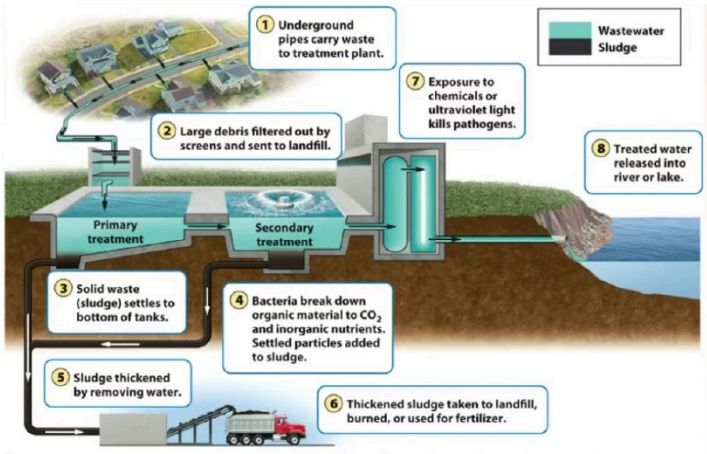
Water Resource

Technology Used to Water Treatment



Water Resource

Sewage Treatment Plant



Water Resource

Sewage Treatment Plant

